# Modulasi Frekuensi (FM)

## ➢ Modulasi Sudut :

- **Definisi:** Modulasi sudut adalah modulasi yang mana sudut sinusoida gelombang sinyal pembawa ($\theta$) diubah-ubah terhadap waktu.

$$M(t) = V_c \, Cos \, [\omega_c t + \theta(t)]$$

Dimana :

$M(t)$ = gelombang sinyal pembawa modulasi sudut

$V_c$ = amplitudo puncak gelombang sinyal pembawa

$\omega_c$ = $2 \pi f_c$,

$f_c$ = frekuensi gelombang sinyal pembawa

$\theta(t)]$ = modulasi sudut [radian]

- Pada modulasi sudut, sinyal *carrier* termodulasi dirumuskan seperti berikut :

$$M(t) = Vc\cos[\mu_c t + \bar{}(t)]$$

dimana Vc dan $\omega_c$ konstan, sementara sudut fasa Φ(t) adalah merupakan fungsi dari sinyal informasi Vm(t)

- Persamaan diatas dapat di kita tulis kembali menjadi persamaan berikut :

$$M(t) = Vc\cos¤(t).$$

Dimana : $¤(t) = \mu_c t + \bar{}(t).$

- **Modulasi sudut :**

  1. Modulasi Fasa (PM = Phase Modulation).
  2. Modulasi Frekuensi (FM = Frequency Modulation)

  - **Modulasi Fasa:** Modulasi sudut yang mana sudut fasa sesaat gelombang sinyal pembawa berubah menyimpang dari harga referensi, sebanding dengan perubahan amplitudo gelombang sinyal pemodulasi (sinyal informasi).

  - **Modulasi Frekuensi :** Modulasi sudut yang mana frekuensi sesaat gelombang sinus sinyal pembawa berubah sebanding dengan harga sesaat (aplitudo sesaat) gelombang sinyal pemodulasi (sinyal informasi).

Sinyal pemodulasi (Informasi), $V_m(t) = V_m \cos \omega_m t$ ,
Sinyal pembawa (carrier), $V_c(t) = V_c \cos \omega_c t$

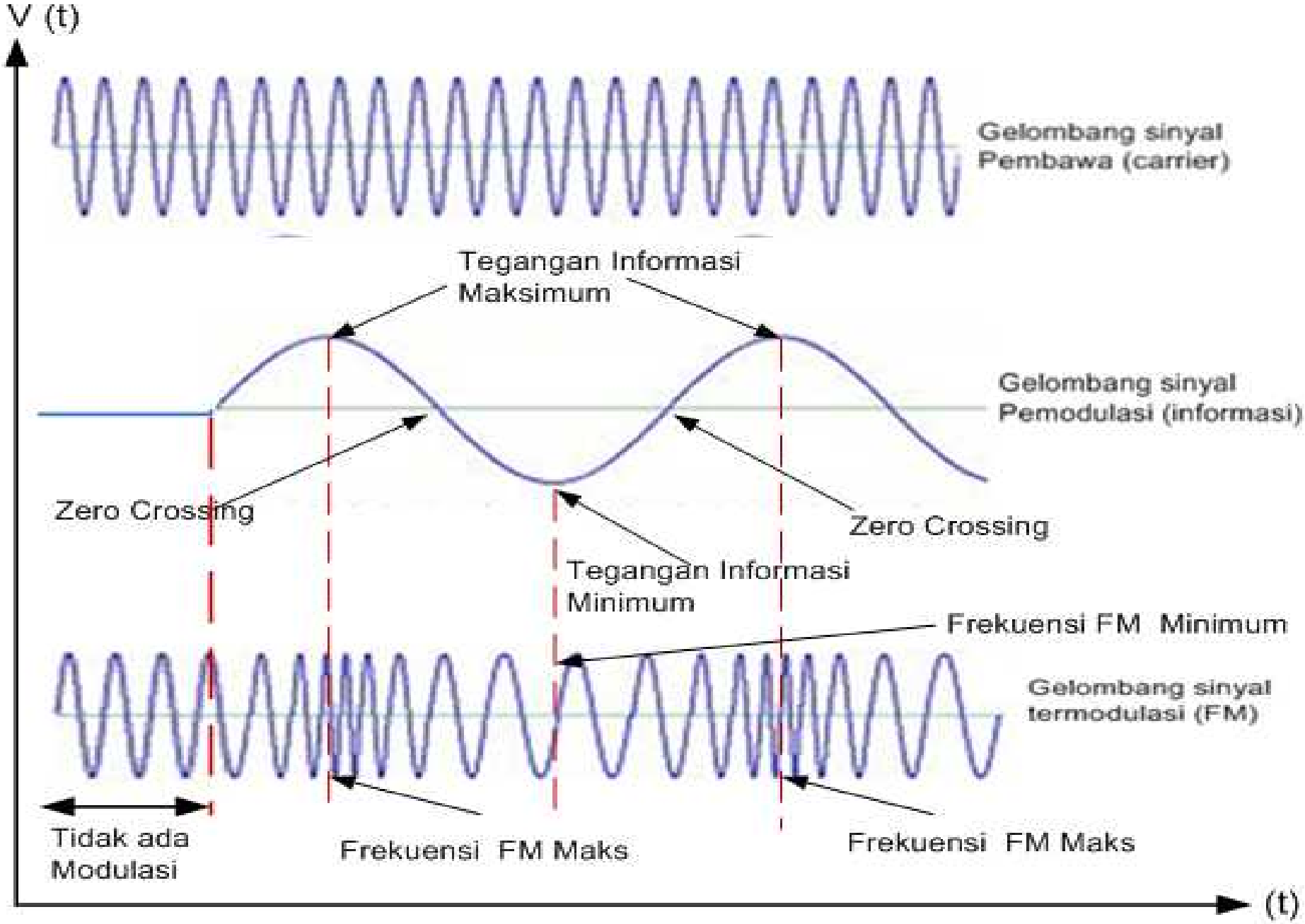
- Persamaan matematis sinyal termodulasi :

$$PM(t) = V_c \cos \omega_c t + k V_m \cos \omega_m t$$

$$FM(t) = V_c \cos [\omega_c t + k_1 V_m / f_m \sin \omega_m t]$$

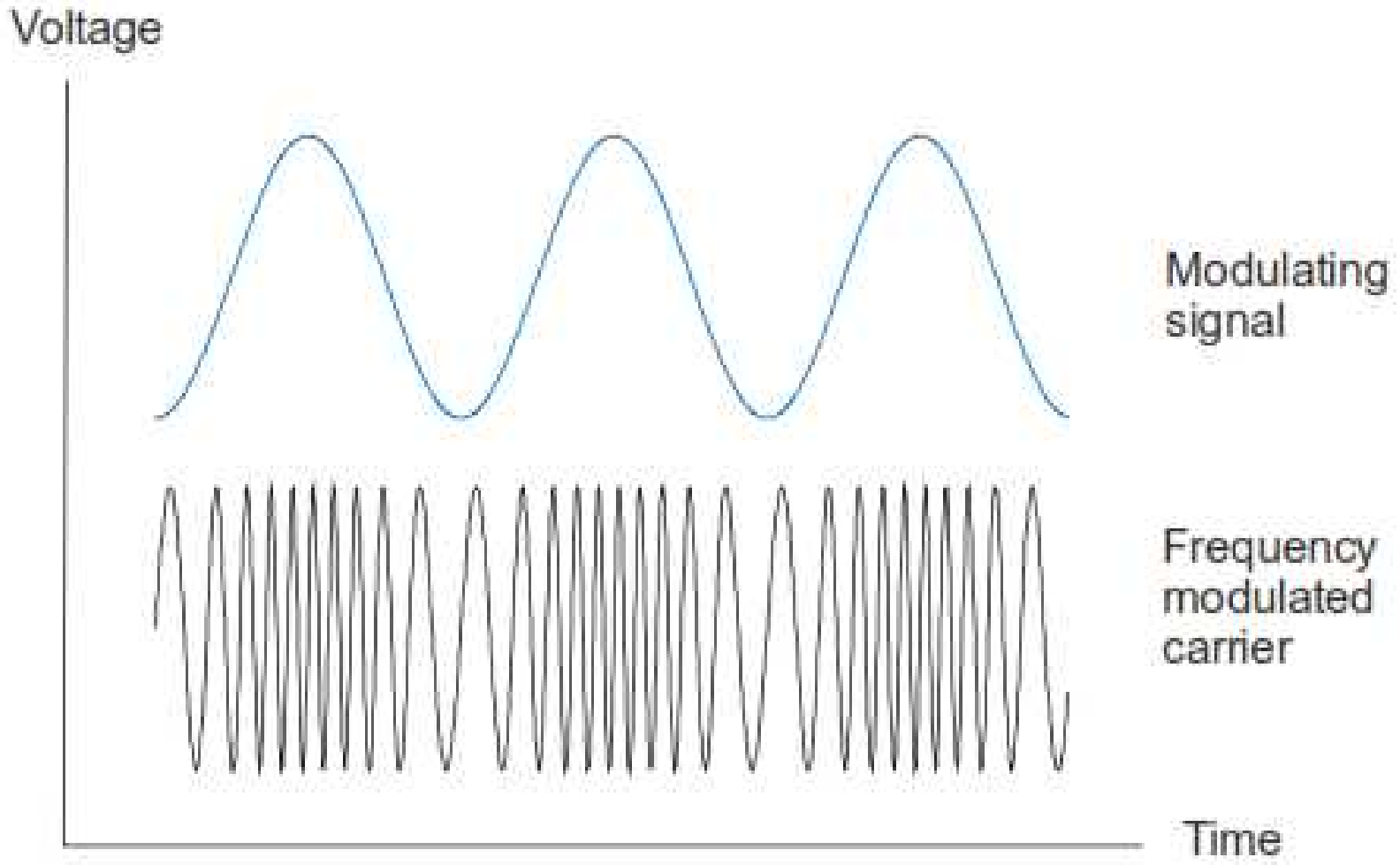
- ❖ Dimana k dan $k_1$ merupakan suatu konstanta dari sensitivitas modulator
- ❖ Untuk modulator PM, k = [radian/volt],
- ❖ Untuk modulator FM, $k_1$ = [Hertz/volt].

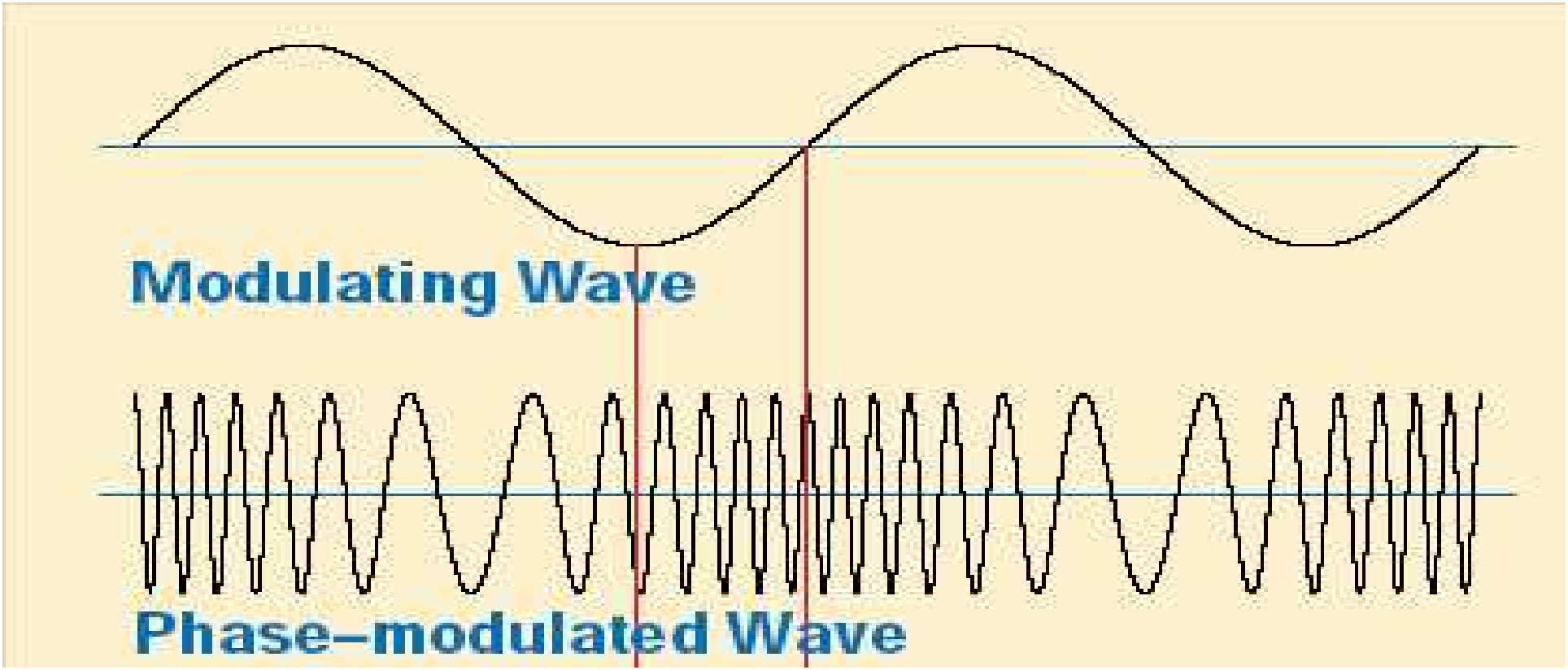
# ➢ Bentuk Gelombang PM dan FM



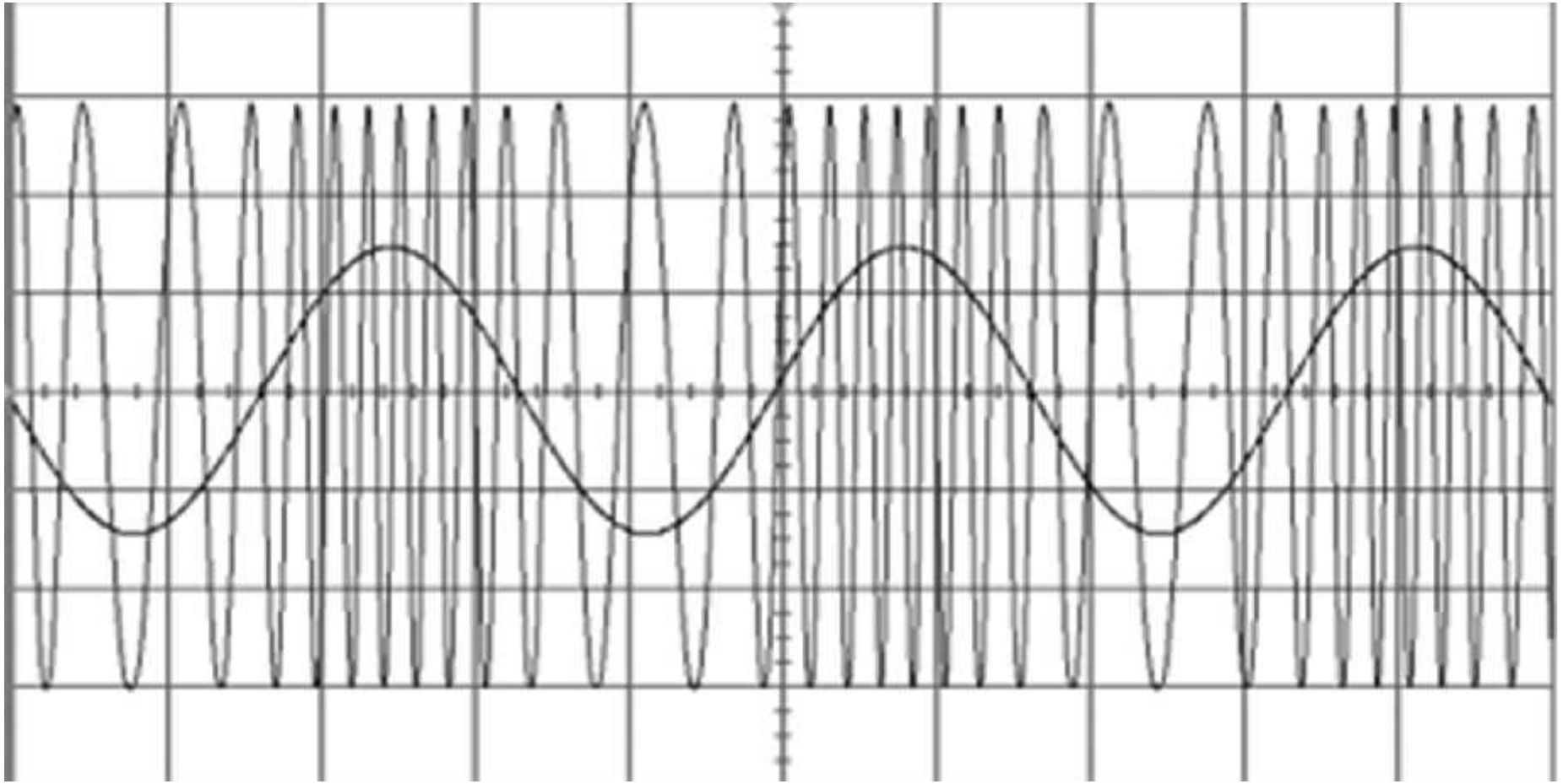
# Frequency modulation

- Frequency is “wobbled” higher and lower by modulating signal

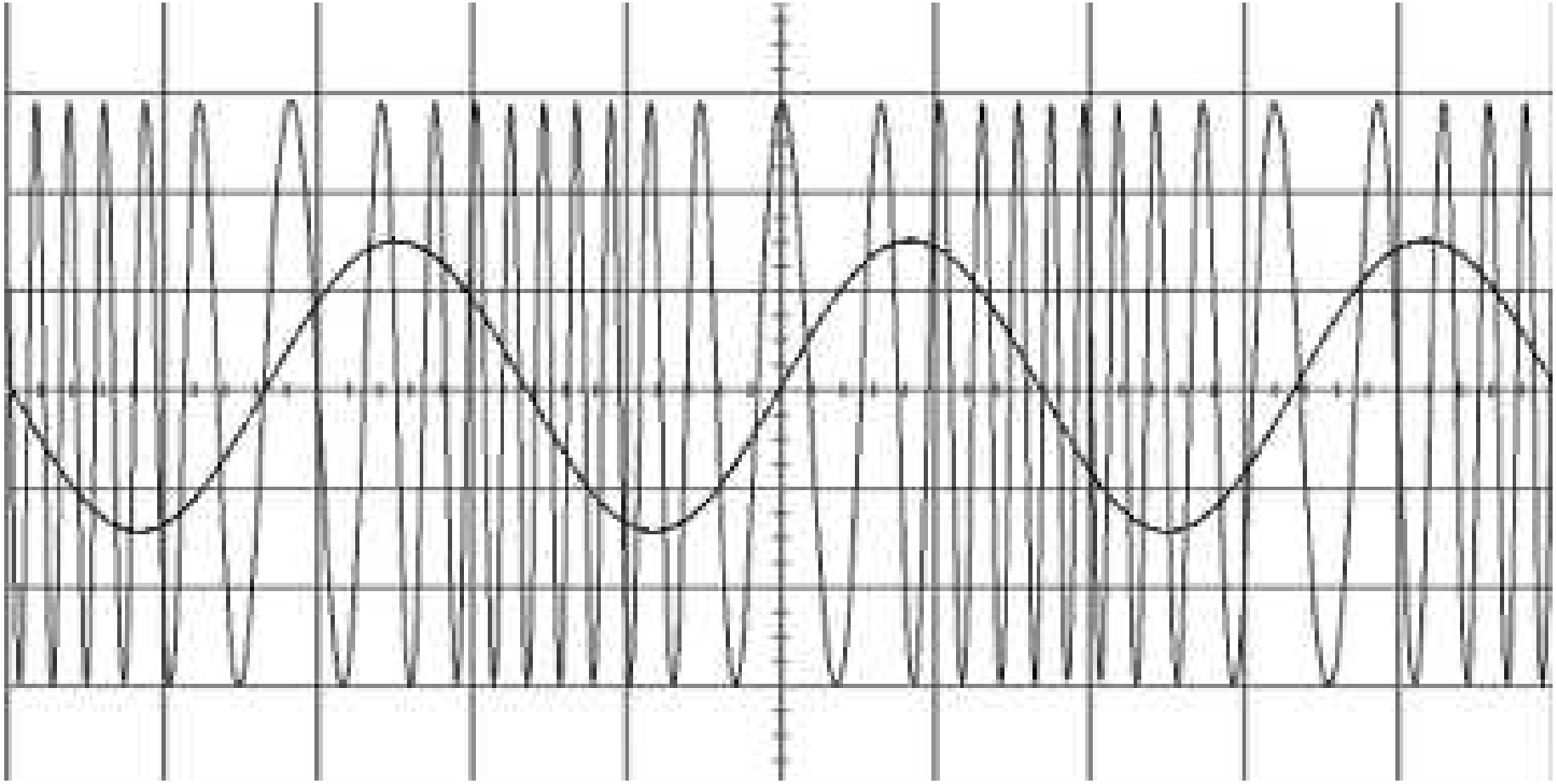



Modulating Wave
Phase–modulated Wave

# Modulasi Frekuensi (FM)

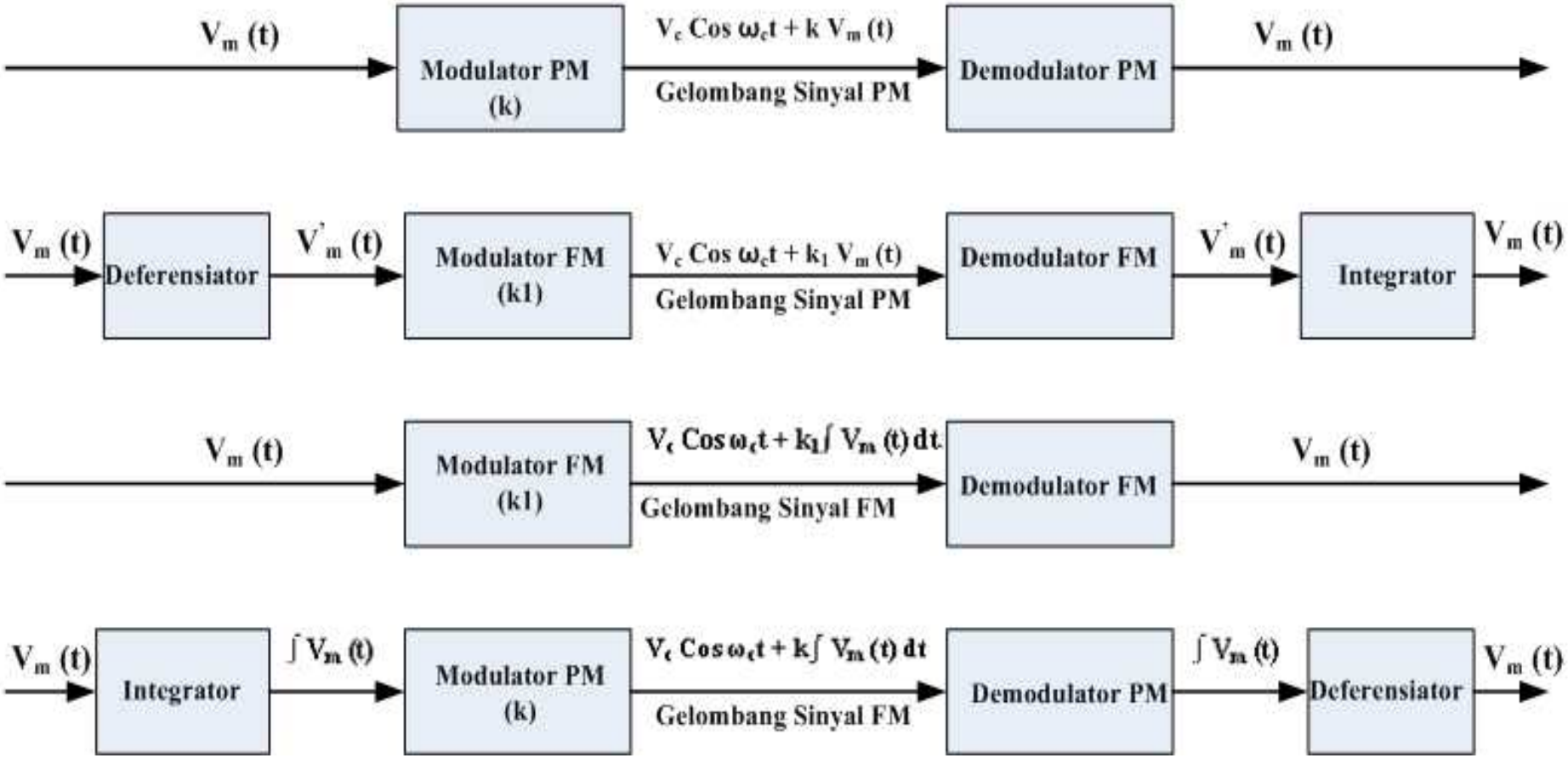


- Modulas PHASA(PM)



# ➢ Modulator dan Demodulator PM, FM

- Modulator PM = Deferensiator + Modulator FM
- Modulator FM = Integrator + Modulator PM
- Demodulator FM = Demodulator PM + Deferensiator.
- Demodulator PM = Demodulator FM + Integrator



## ➢ Analisis Sinyal FM

- Sinyal pemodulasi $Vm(t) = V_m \, Cos \, \omega_m t$
- Sinyal pembawa $Vc(t) = V_c \, Cos \, \omega_c t$,
- Sinyal FM :

$$FM(t) = V_c \, Cos \, [\omega_c t + m \, Cos \, \omega_m t]$$

Dimana m = Indeks modulasi

$$m = \frac{\Delta f}{fm}$$

Δf = Simpangan / deviasi maksimum sinyal FM

Fm = frekuensi maksimum sinyal informasi

- Dengan fungsi Bessel bahwa :

$$\text{Cos}\,(\alpha + \text{m Cos}\,\beta) = \sum_{n=-x}^{x} \text{Jn(m) Cos}\,(\alpha + n\beta + n\pi/2)$$

$$\text{Maka : } \text{FM(t)} = V_c \text{ Cos } [\omega_c t + \text{m Cos } \omega_m t]$$

$$\textbf{FM(t) :} \{[J_0(m) \text{ Cos } \omega_c t + J_1(m) \text{ Cos } ((\omega_c - \omega_m)\, t$$

$$+ \pi/2] + J_1(m) \text{ Cos } [(\omega_c + \omega_m)\, t + \pi/2]$$

$$- J_2(m) \text{ Cos } [(\omega_c + 2\omega_m)\, t] - J_2(m) \text{ Coc } [(\omega_c + 2\omega_m)\, t]$$

+.....dst

Dimana :

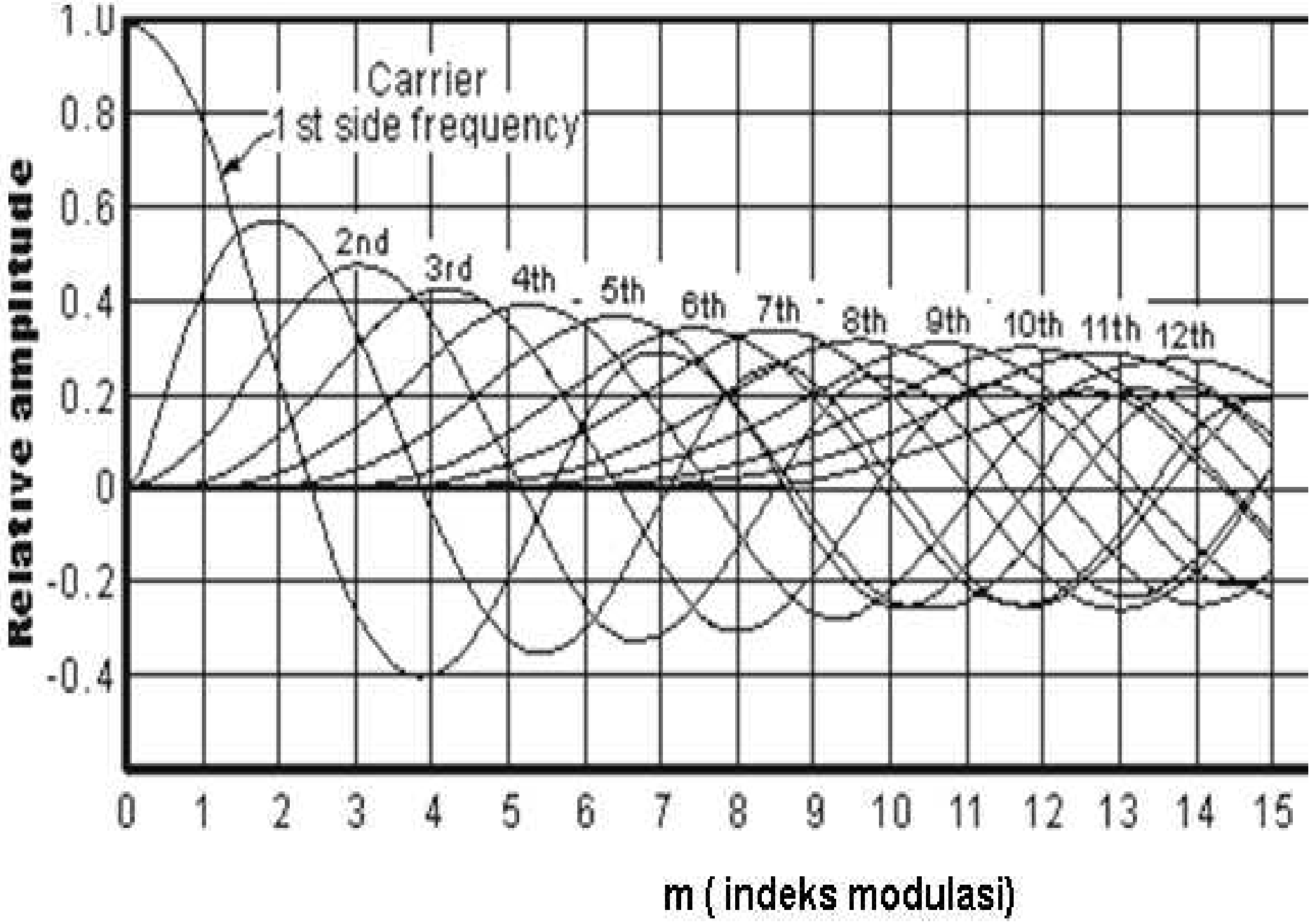
- m = indeks modulasi
- $J_0$ = % amplitudo gelombang sinyal pembawa (carrier) termodulasi terhadap amplitudo gelombang sinyal carrier saat tidak terjadi modulasi.
- $J_1$ = % amplitudo frekuensi-frekuensi sisi orde pertama terhadap amplitudo gelombang sinyal carrier saat tidak terjadi modulasi.
- $J_2$ = % amplitudo frekuensi-frekuensi sisi orde kedua terhadap amplitudo gelombang sinyal carrier saat tidak terjadi modulasi
- $J_3$ = % amplitudo frekuensi-frekuensi sisi orde ketiga terhadap amplitudo gelombang sinyal carrier saat tidak terjadi modulasi
- $J_4$ = % amplitudo frekuensi-frekuensi sisi orde keempat terhadap amplitudo gelombang sinyal carrier saat tidak terjadi modulasi

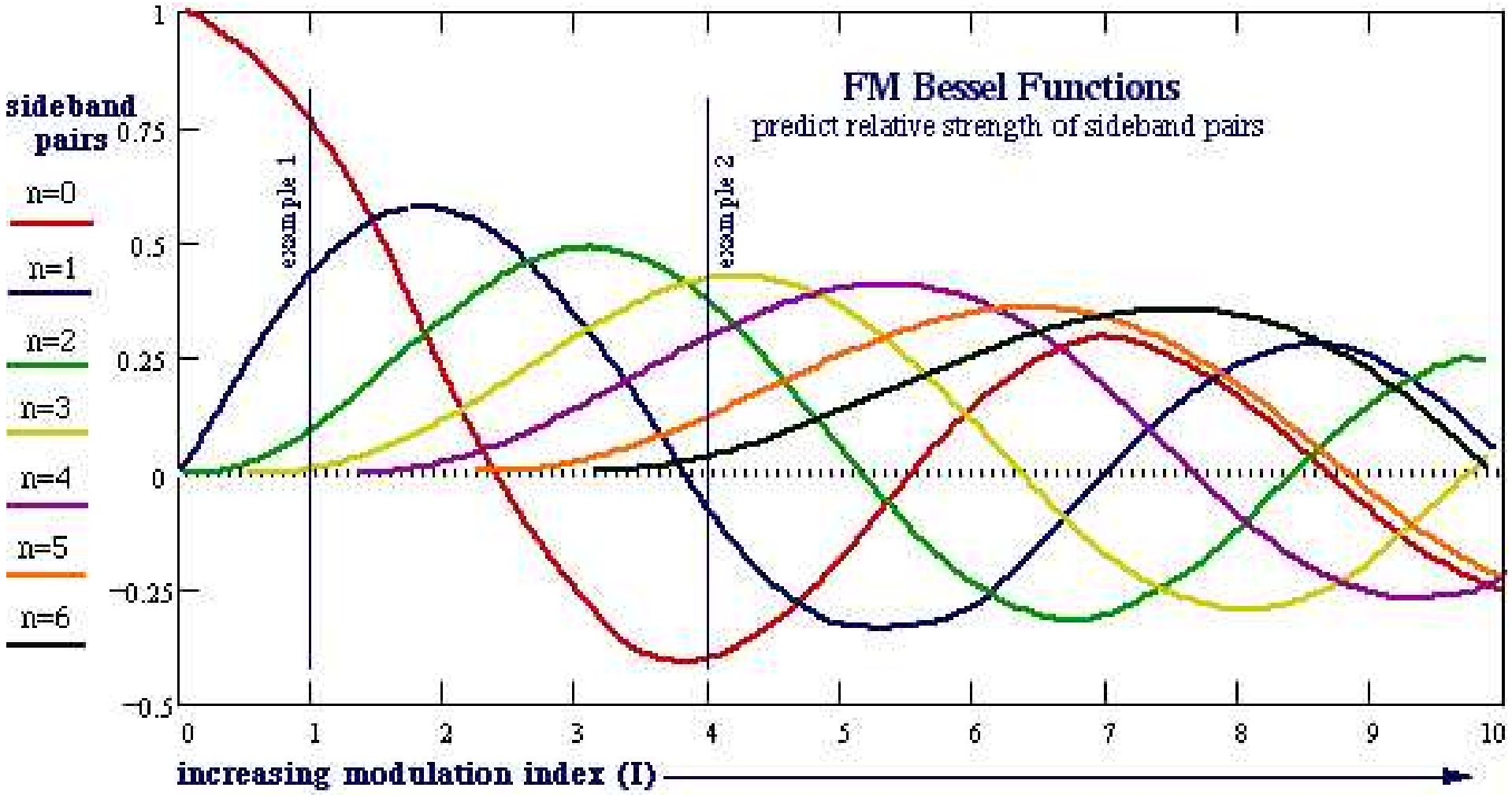
Dan seterusnya

# Frekuensi-frekuensi Sisi Gelombang Sinyal FM dari Fungsi Bessel

| $x$ (CARRIER) | // OR ORDER | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ($m_f$) | $J_0$ | $J_1$ | $J_2$ | $J_3$ | $J_4$ | $J_5$ | $J_6$ | $J_7$ | $J_8$ | $J_9$ | $J_{10}$ | $J_{11}$ | $J_{12}$ | $J_{13}$ | $J_{14}$ | $J_{15}$ | $J_{16}$ |
| 0.00 | 1.00 | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 0.25 | 0.98 | 0.12 | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 0.5 | 0.94 | 0.24 | 0.03 | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 1.0 | 0.77 | 0.44 | 0.11 | 0.02 | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 1.5 | 0.51 | 0.56 | 0.23 | 0.06 | 0.01 | — | — | — | — | — | — | — | — | — | — | — | — |
| 2.0 | 0.22 | 0.58 | 0.35 | 0.13 | 0.03 | — | — | — | — | — | — | — | — | — | — | — | — |
| 2.5 | −0.05 | 0.50 | 0.45 | 0.22 | 0.07 | 0.02 | — | — | — | — | — | — | — | — | — | — | — |
| 3.0 | −0.26 | 0.34 | 0.49 | 0.31 | 0.13 | 0.04 | 0.01 | — | — | — | — | — | — | — | — | — | — |
| 4.0 | −0.40 | −0.07 | 0.36 | 0.43 | 0.28 | 0.13 | 0.05 | 0.02 | — | — | — | — | — | — | — | — | — |
| 5.0 | −0.18 | −0.33 | 0.05 | 0.36 | 0.39 | 0.26 | 0.13 | 0.05 | 0.02 | — | — | — | — | — | — | — | — |
| 6.0 | 0.15 | −0.28 | −0.24 | 0.11 | 0.36 | 0.36 | 0.25 | 0.13 | 0.06 | 0.02 | — | — | — | — | — | — | — |
| 7.0 | 0.30 | 0.00 | −0.30 | −0.17 | 0.16 | 0.35 | 0.34 | 0.23 | 0.13 | 0.06 | 0.02 | — | — | — | — | — | — |
| 8.0 | 0.17 | 0.23 | −0.11 | −0.29 | −0.10 | 0.19 | 0.34 | 0.32 | 0.22 | 0.13 | 0.06 | 0.03 | — | — | — | — | — |
| 9.0 | −0.09 | 0.24 | 0.14 | −0.18 | −0.27 | −0.06 | 0.20 | 0.33 | 0.30 | 0.21 | 0.12 | 0.06 | 0.03 | 0.01 | — | — | — |
| 10.0 | −0.25 | 0.04 | 0.25 | 0.06 | −0.22 | −0.23 | −0.01 | 0.22 | 0.31 | 0.29 | 0.20 | 0.12 | 0.06 | 0.03 | 0.01 | — | — |
| 12.0 | 0.05 | −0.22 | −0.08 | 0.20 | 0.18 | −0.07 | −0.24 | −0.17 | 0.05 | 0.23 | 0.30 | 0.27 | 0.20 | 0.12 | 0.07 | 0.03 | 0.01 |
| 15.0 | −0.01 | 0.21 | 0.04 | −0.19 | −0.12 | 0.13 | 0.21 | 0.03 | −0.17 | −0.22 | −0.09 | 0.10 | 0.24 | 0.28 | 0.25 | 0.18 | 0.12 |

Source: E. Cambi, *Bessel Functions*, Dover Publications, Inc., New York, 1948. Courtesy of the publisher.


Relative amplitude
1.U
0.8
0.6
0.4
0.2
0
-0.2
-0.4
Carrier
1 st side frequency
2nd
3rd
4th
5th
6th
7th
8th
9th
10th
11th
12th
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
m ( indeks modulasi)


FM Bessel Functions
predict relative strength of sideband pairs
sideband pairs
n=0
n=1
n=2
n=3
n=4
n=5
n=6
example 1
example 2
1
0.75
0.5
0.25
0
−0.25
−0.5
0
1
2
3
4
5
6
7
8
9
10
increasing modulation index (I)

$$v_{angle}(t) = V_c \underbrace{\cos\left(\mu_c t + m\cos(\mu_m t)\right)}$$

$$\cos(\sim + m\cos\quad) = \sum_{n=-\infty}^{\infty} J_n(m)\cos\left(\sim + n\quad + \frac{n®}{2}\right)$$

**Jn (m) is the Bessel function of the first kind**

$$v_{angle}(t) = V_c \sum_{n=-\infty}^{\infty} J_n(m)\cos\left(\mu_c t + n\mu_m t + \frac{n®}{2}\right)$$

$$v_{angle}(t) = V_c \left\{ \begin{array}{l} J_0(m)\cos\mu_c t + \\ J_1(m)\cos\left[(\mu_c + \mu_m)t + \frac{®}{2}\right] - J_1(m)\cos\left[(\mu_c - \mu_m)t - \frac{®}{2}\right] + \\ J_2(m)\cos\left[(\mu_c + 2\mu_m)t + ®\right] + J_2(m)\cos\left[(\mu_c - 2\mu_m)t - ®\right] + ..... \end{array} \right\}$$

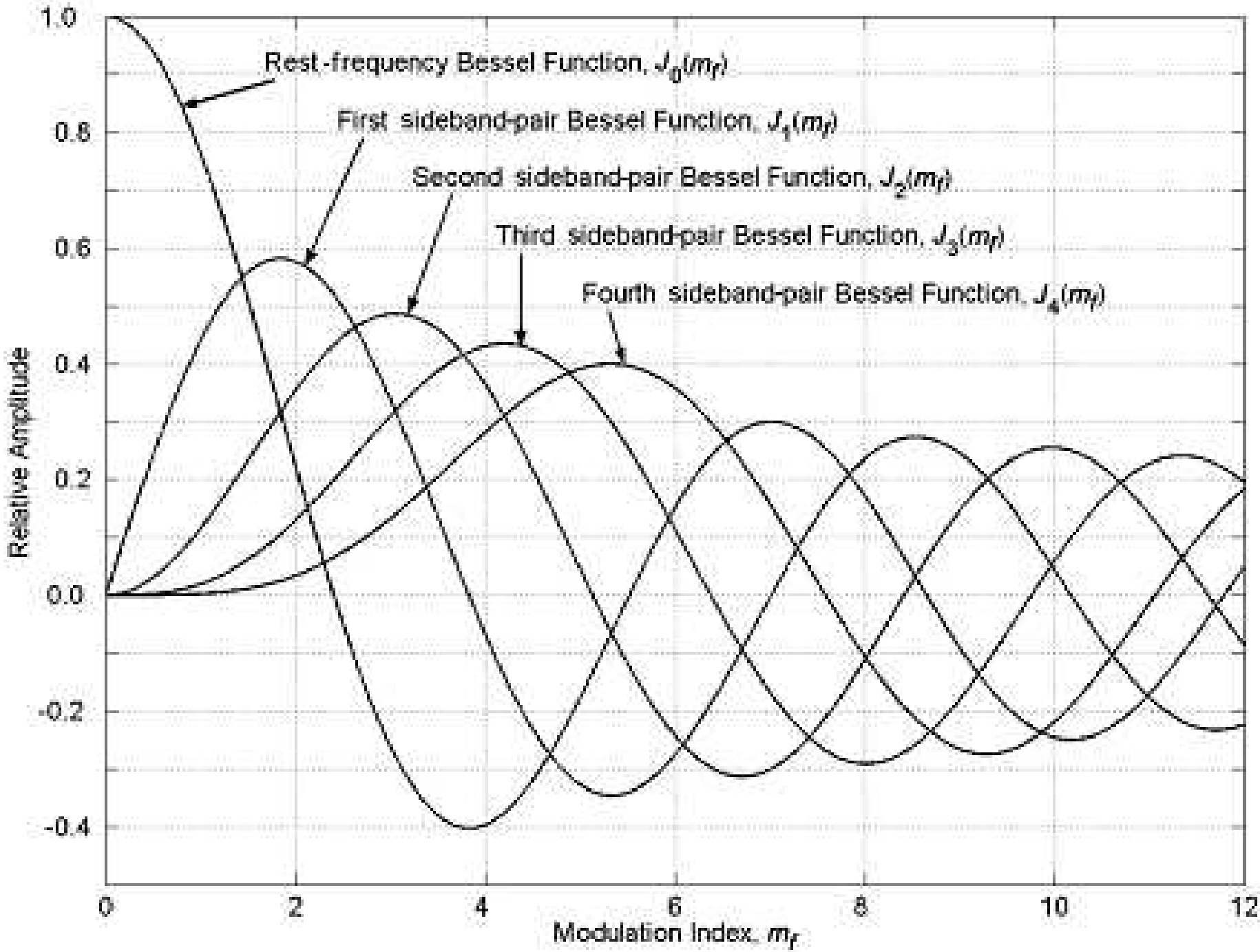
**FM:** $m = \dfrac{K_f V_m}{\mu_m}$ **PM:** $m = K_p V_m$

- Fungsi Bessel yg berkaitan dg FM

$$e_{FM}(t) = E_C \begin{cases} J_0(m_f)\sin\omega_C t - J_1(m_f)\left[\sin(\omega_C - \omega_M)t - \sin(\omega_C + \omega_M)t\right] + \\ J_2(m_f)\left[\sin(\omega_C - 2\omega_M)t + \sin(\omega_C + 2\omega_M)t\right] - \\ J_3(m_f)\left[\sin(\omega_C - 3\omega_M)t - \sin(\omega_C + 3\omega_M)t\right] + \\ J_4(m_f)\left[\sin(\omega_C - 4\omega_M)t + \sin(\omega_C + 4\omega_M)t\right] - \ldots \end{cases}$$

# ➢Tabel Fungsi Bessel

| $x$ | Bessel-function order, n | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $m_t$ | $J_0$ | $J_1$ | $J_2$ | $J_3$ | $J_4$ | $J_5$ | $J_6$ | $J_7$ | $J_8$ | $J_9$ | $J_{10}$ | $J_{11}$ | $J_{12}$ | $J_{13}$ | $J_{14}$ | $J_{15}$ | $J_{16}$ |
| 0.00 | 1.00 | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 0.25 | 0.98 | 0.12 | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 0.5 | 0.94 | 0.24 | 0.03 | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 1.0 | 0.77 | 0.44 | 0.11 | 0.02 | — | — | — | — | — | — | — | — | — | — | — | — | — |
| 1.5 | 0.51 | 0.56 | 0.23 | 0.06 | 0.01 | — | — | — | — | — | — | — | — | — | — | — | — |
| 2.0 | 0.22 | 0.58 | 0.35 | 0.13 | 0.03 | — | — | — | — | — | — | — | — | — | — | — | — |
| 2.41 | 0 | 0.52 | 0.43 | 0.20 | 0.06 | 0.02 | — | — | — | — | — | — | — | — | — | — | — |
| 2.5 | −.05 | 0.50 | 0.45 | 0.22 | 0.07 | 0.02 | 0.01 | — | — | — | — | — | — | — | — | — | — |
| 3.0 | −.26 | 0.34 | 0.49 | 0.31 | 0.13 | 0.04 | 0.01 | — | — | — | — | — | — | — | — | — | — |
| 4.0 | −.40 | −.07 | 0.36 | 0.43 | 0.28 | 0.13 | 0.05 | 0.02 | — | — | — | — | — | — | — | — | — |
| 5.0 | −.18 | −.33 | 0.05 | 0.36 | 0.39 | 0.26 | 0.13 | 0.05 | 0.02 | — | — | — | — | — | — | — | — |
| 5.53 | 0 | −.34 | −.13 | 0.25 | 0.40 | 0.32 | 0.19 | 0.09 | 0.03 | 0.01 | — | — | — | — | — | — | — |
| 6.0 | 0.15 | −.28 | −.24 | 0.11 | 0.36 | 0.36 | 0.25 | 0.13 | 0.06 | 0.02 | — | — | — | — | — | — | — |
| 7.0 | 0.30 | 0.00 | −.30 | −.17 | 0.16 | 0.35 | 0.34 | 0.23 | 0.13 | 0.06 | 0.02 | — | — | — | — | — | — |
| 8.0 | 0.17 | 0.23 | −.11 | −.29 | −.10 | 0.19 | 0.34 | 0.32 | 0.22 | 0.13 | 0.06 | 0.03 | — | — | — | — | — |
| 8.65 | 0 | 0.27 | 0.06 | −.24 | −.23 | 0.03 | 0.26 | 0.34 | 0.28 | 0.18 | 0.10 | 0.05 | 0.02 | — | — | — | — |
| 9.0 | −.09 | 0.25 | 0.14 | −.18 | −.27 | −.06 | 0.20 | 0.33 | 0.31 | 0.21 | 0.12 | 0.06 | 0.03 | 0.01 | — | — | — |
| 10.0 | −.25 | 0.04 | 0.25 | 0.06 | −.22 | −.23 | −.01 | 0.22 | 0.32 | 0.29 | 0.21 | 0.12 | 0.06 | 0.03 | 0.01 | — | — |
| 12.0 | 0.05 | −.22 | −.08 | 0.20 | 0.18 | −.07 | −.24 | −.17 | 0.05 | 0.23 | 0.30 | 0.27 | 0.20 | 0.12 | 0.07 | 0.03 | 0.01 |


Rest-frequency Bessel Function, $J_0(m_f)$
First sideband-pair Bessel Function, $J_1(m_f)$
Second sideband-pair Bessel Function, $J_2(m_f)$
Third sideband-pair Bessel Function, $J_3(m_f)$
Fourth sideband-pair Bessel Function, $J_4(m_f)$
Relative Amplitude
1.0
0.8
0.6
0.4
0.2
0.0
-0.2
-0.4
0
2
4
6
8
10
12
Modulation Index, $m_f$

## ➢ Spektrum dan Bandwidth Sinyal FM

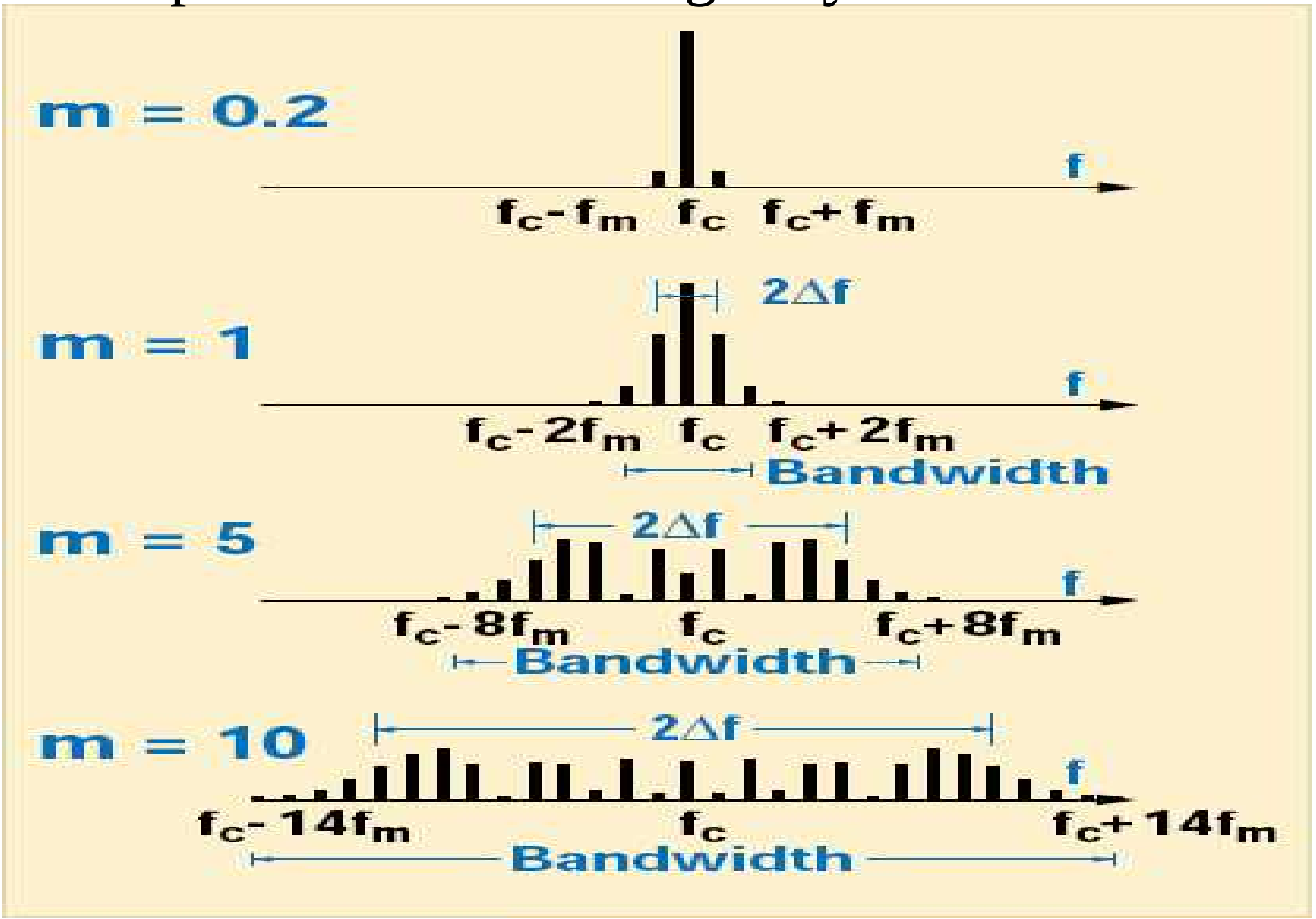
❖ Berdasarkan fungsi Bessel, bandwidth gelombang sinyal FM :

$$Bw = 2.n.f_m$$

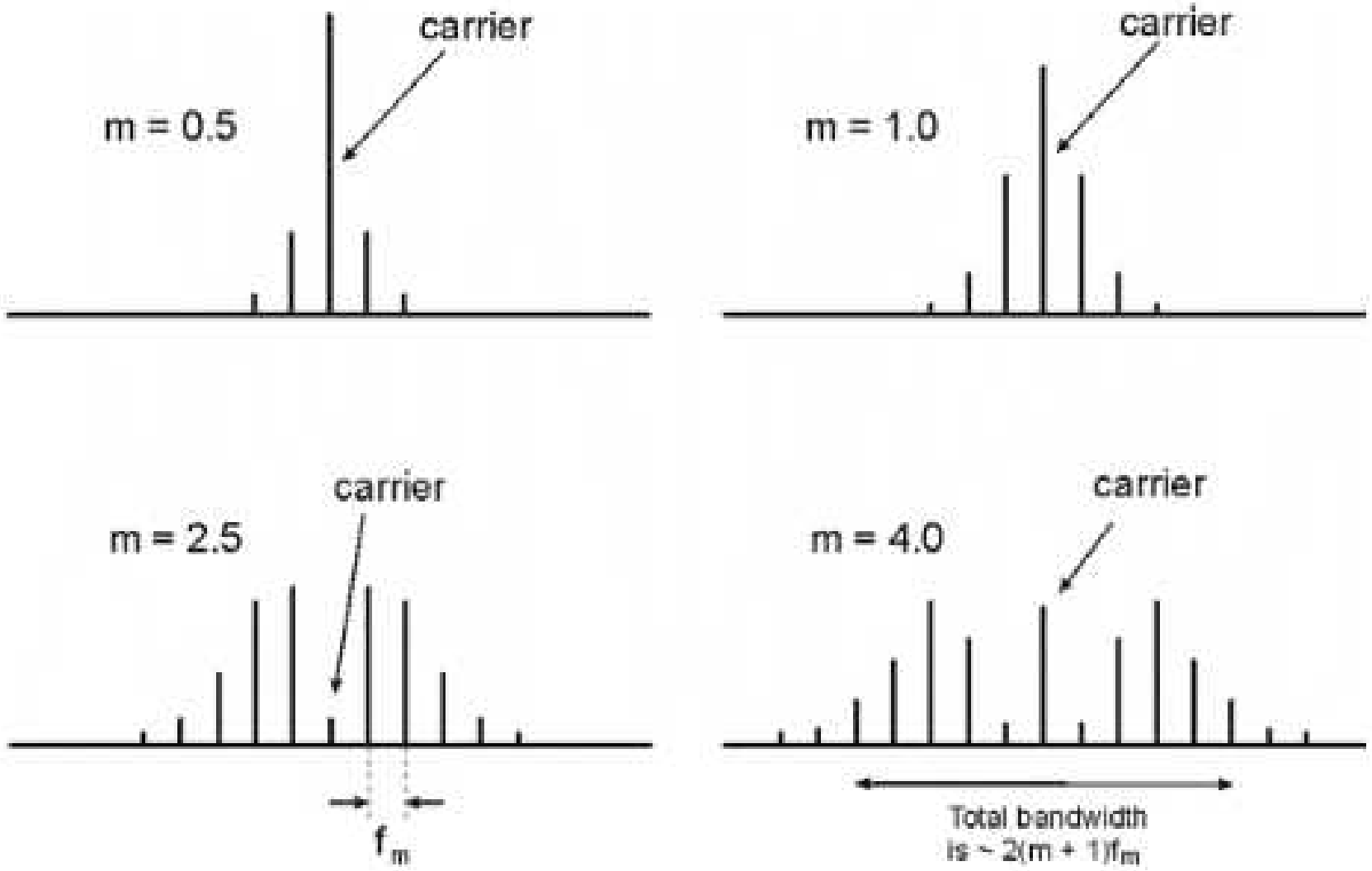
❖ Bandwidth gelombang sinyal FM berdasar atas rumus Carlson :

$$Bw = 2\,(\Delta \mathbf{f} + \mathbf{f_m})$$

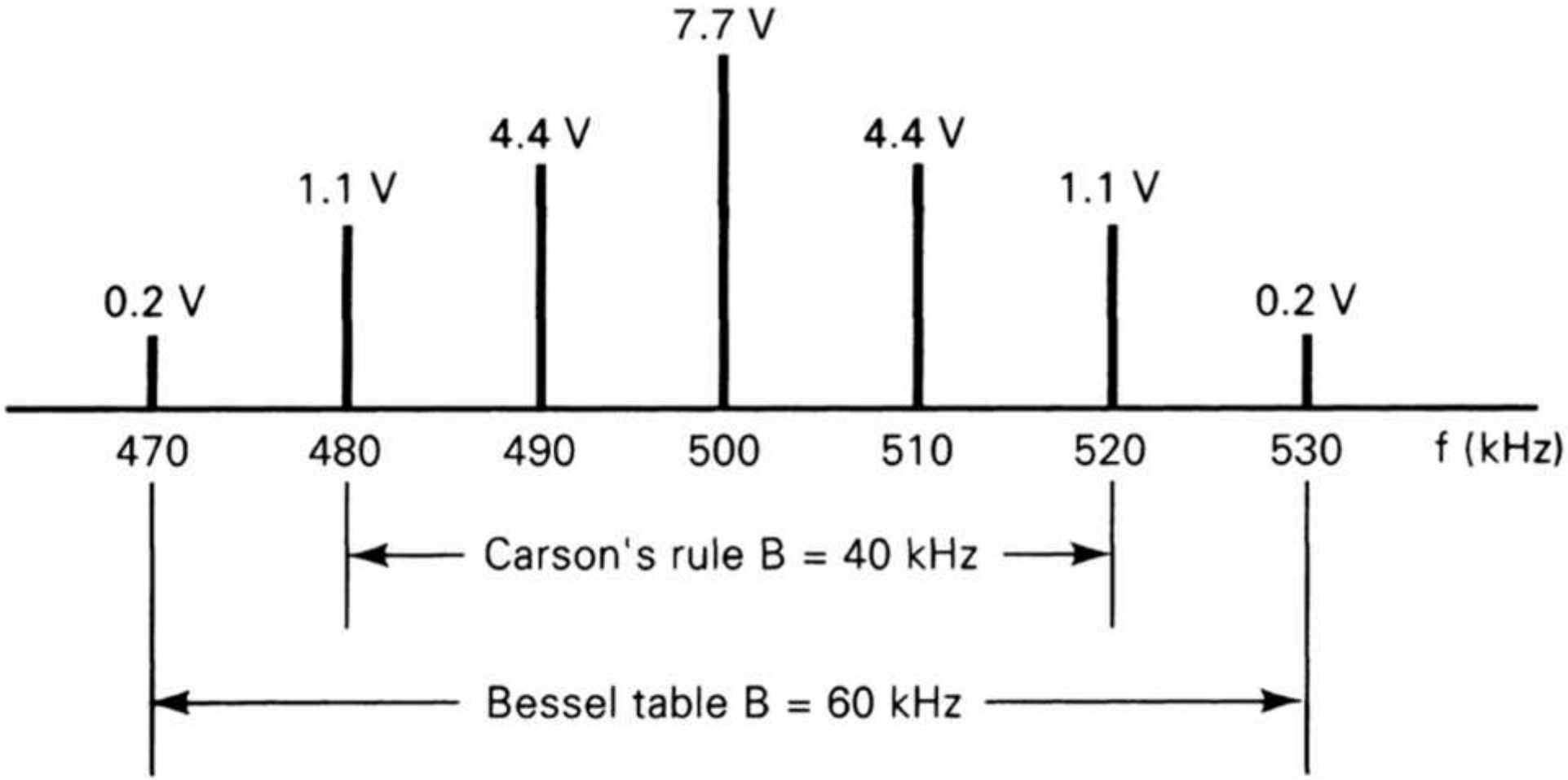
❖ Spektrum Gelombang Sinyal FM :



# FM spectrum


carrier
m = 0.5
carrier
m = 1.0
carrier
m = 2.5
$f_m$
carrier
m = 4.0
Total bandwidth
is ~ 2(m + 1)$f_m$

**FM modulator: af = 10 kHz, $f_m$ = 10 kHz, $V_c$ = 10 V, $f_c$ = 500 kHz, m=1**



- The Index of Modulation

$$m = \frac{\Delta f}{fm}$$

$$m = K_1 V_m / f_m$$

Dimana :

$m$ = indeks modulasi

$\Delta f$ = Deviasi frekuensi maksimum

$fm$ = Frekuensi maksimum sinyal informasi/ sinyal pemodulasi

- Percentage of Modulation

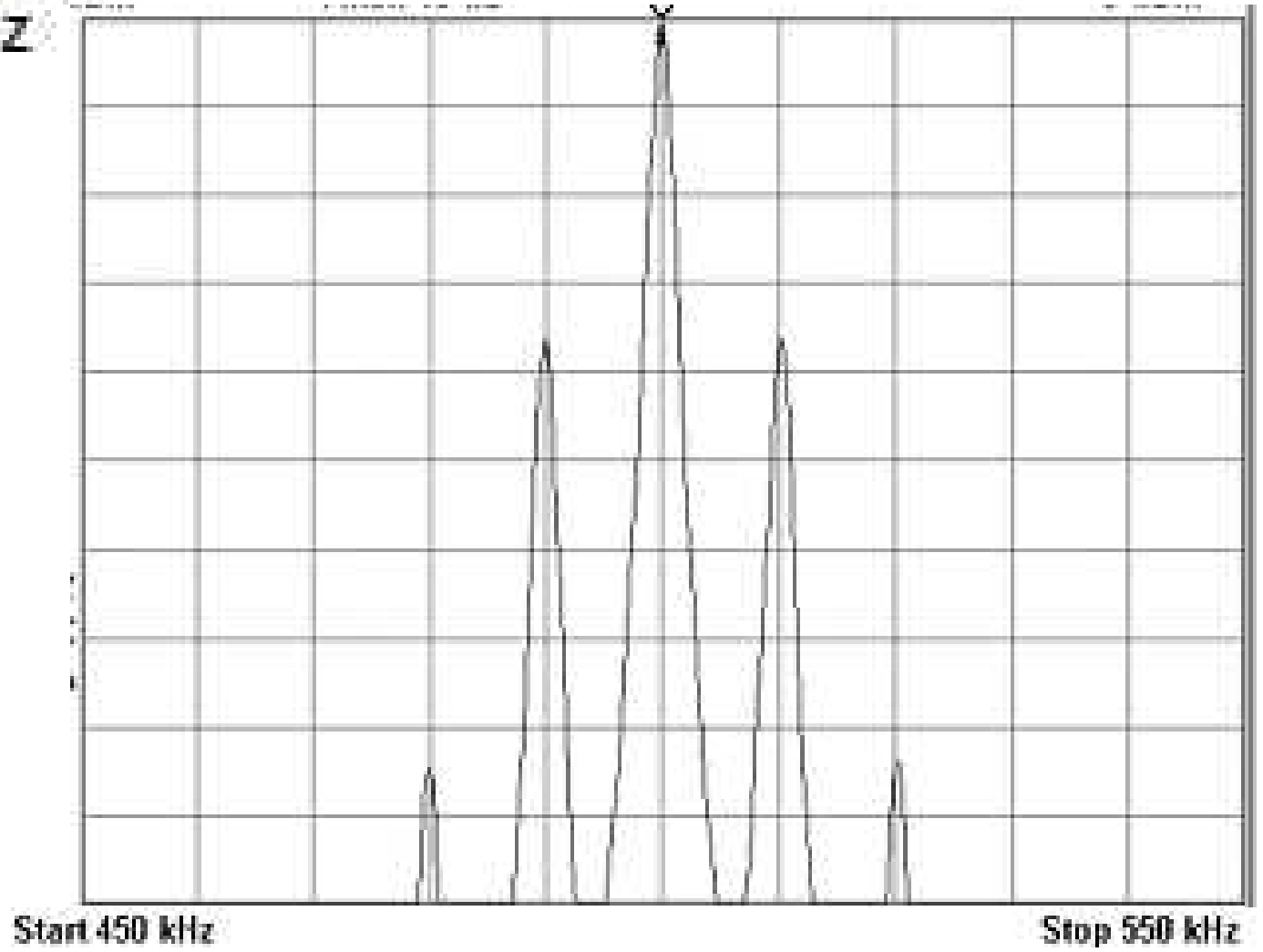
$$\%m = \frac{\Delta f}{f\ maks} \times 100\ \%$$

Dimana : $\Delta f\ maks$ = deviasi maksimum

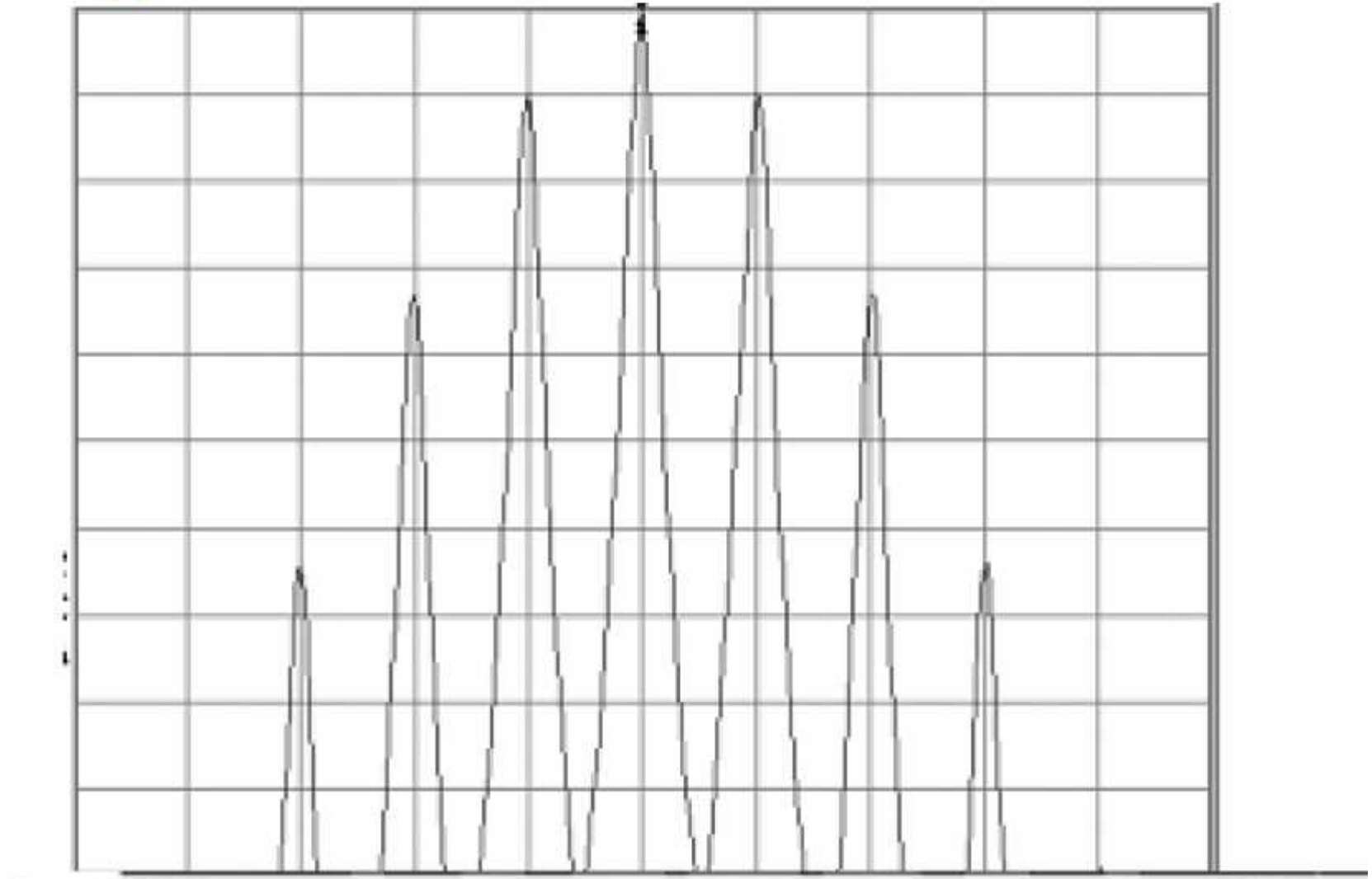
standar siaran FM komersial = 75 Khz

# ➢ Spektrum FM

$m_f = 0.25$ and
$f_M = 10$ kHz



$m_f = 1$ and
$f_M = 10$ kHz

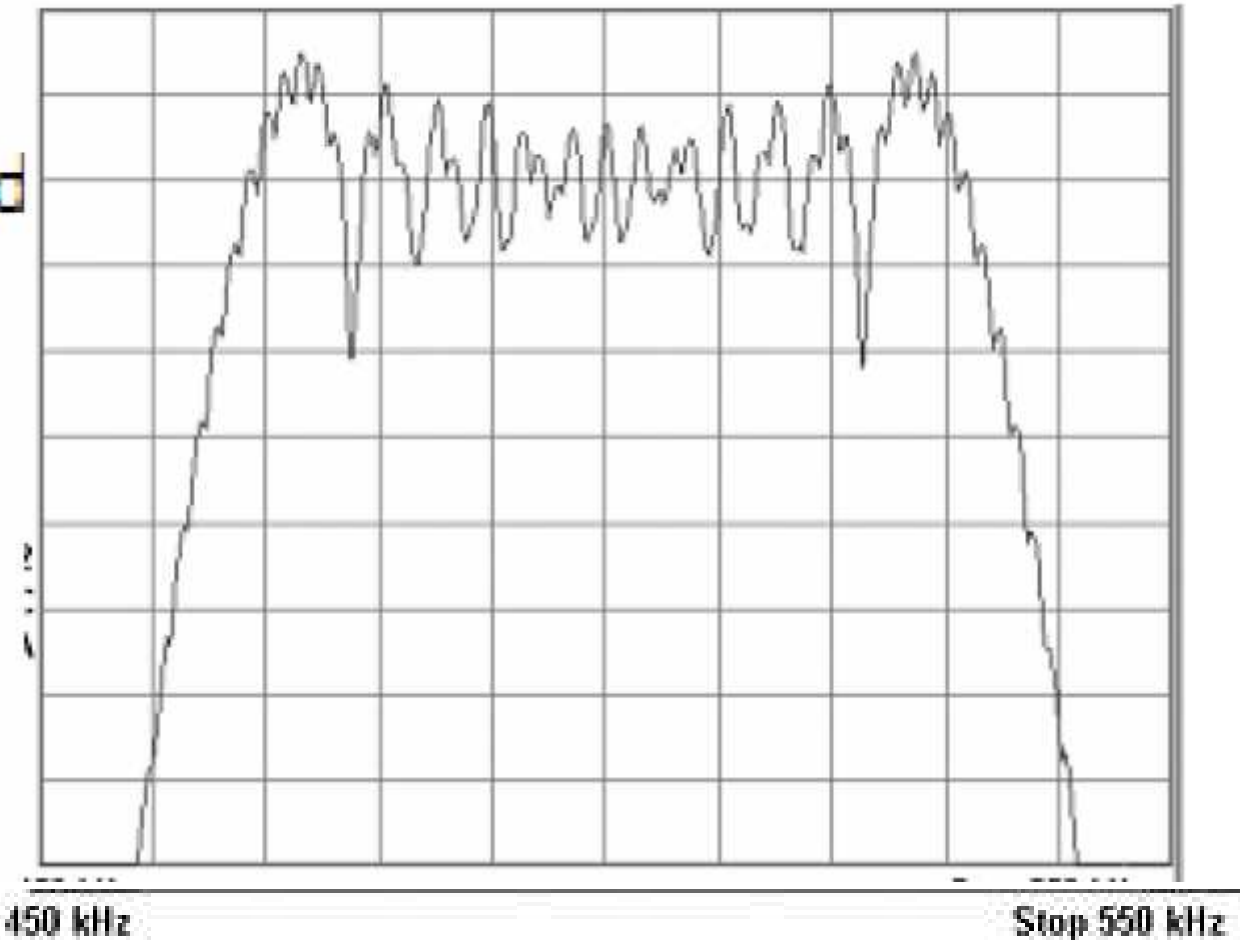


Start 450 kHz

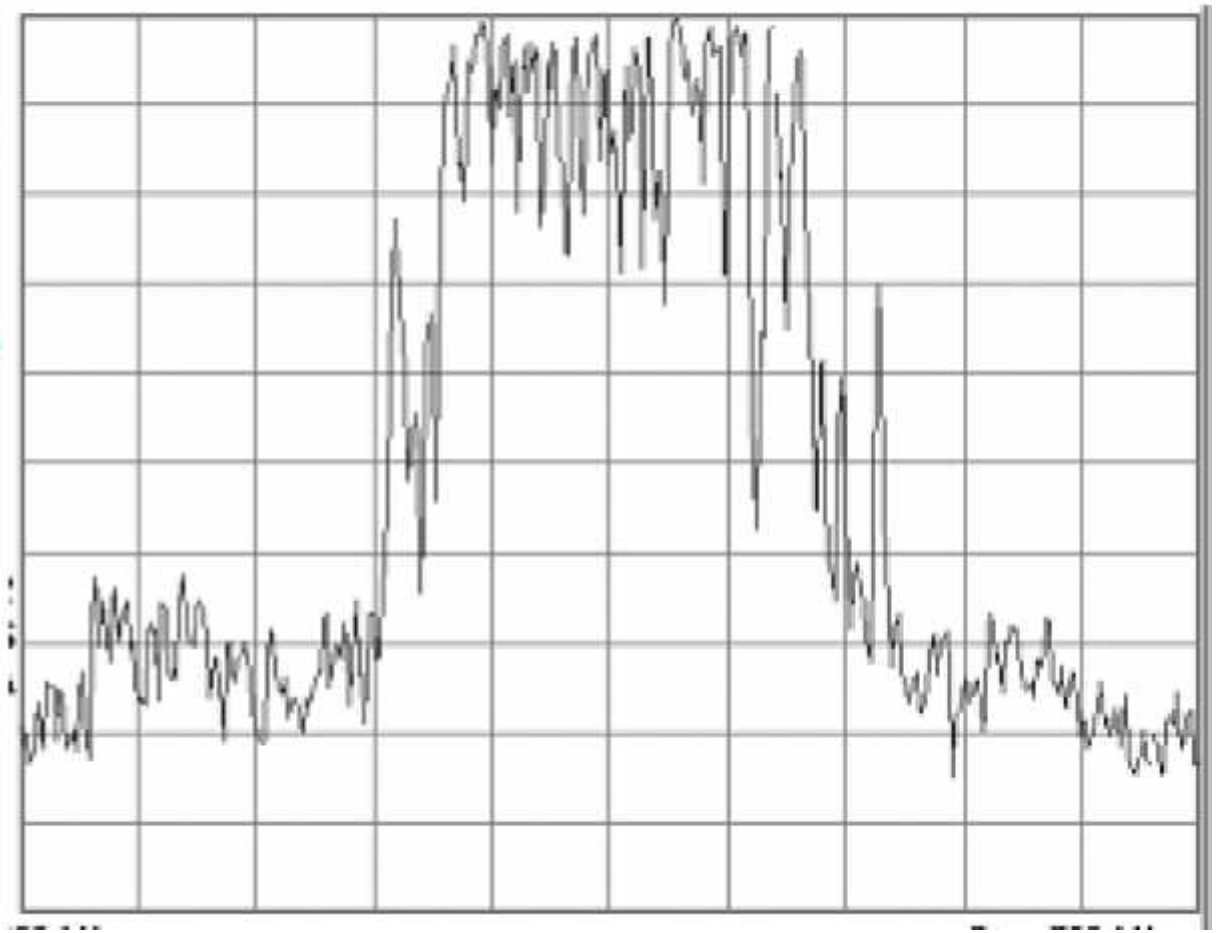
Stop 550 kHz

# Wideband FM

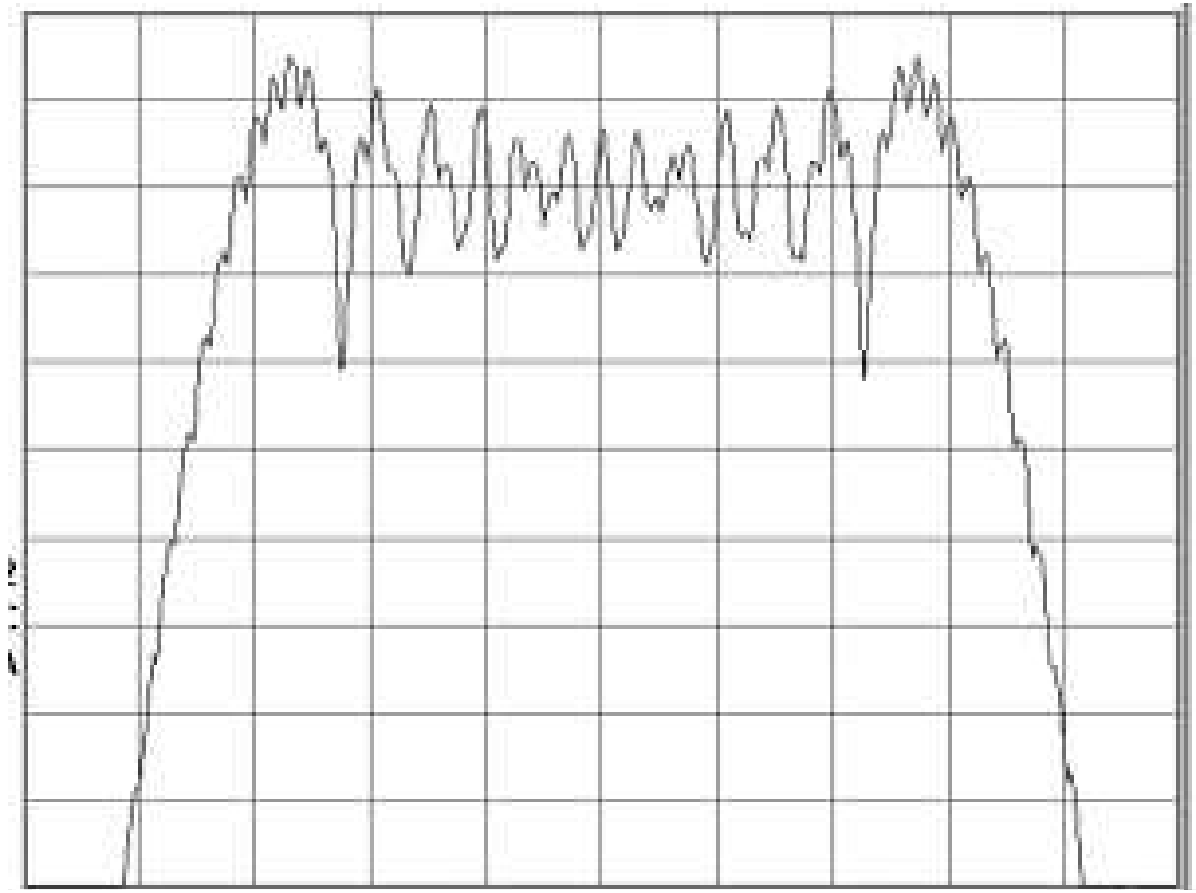
an FM signal
with $m_f = 20$,
$\delta = 30$ kHz, and
$f_M = 1.5$ kHz



An FM signal spectrum. The modulating signal is music.



An FM wave with $m_f \cong 2.405$. Note the null at the center rest frequency.



## ➢ Daya pada Sinyal FM

- Daya yang dipancarkan konstan baik saat tidak terjadi proses modulasi maupun saat terjadi proses modulasi.

$$P_{FM\ tot.} = \frac{(V_c \text{ termodulasi})^2}{R} + 2\frac{(V_{SF1})^2}{R} + 2\frac{(V_{SF2})^2}{R} + 2\frac{(V_{SF3})^2}{R} + \dots + 2\frac{(V_{SFn})^2}{R}$$

# Example

1. Commercial FM broadcasting stations restrict the maximum carrier frequency deviation to 75 KHz . A single modulating tone of 15 KHz is used to produce the peak frequency deviation of the carrier.
   Determine the modulation index ?

$$m = \frac{\Delta f}{f_m}$$

$$m = \frac{75KHz}{15KHz} = 5$$

2. If the peak frequency deviation of the carrier is 100 kHz, and is produced by a 5 kHz modulating tone.
   Determine the modulation index ?

$$m = \frac{\Delta f}{f_m} \qquad m = \frac{100KHz}{5KHz} = 20$$

3. A FM system has a maximum frequency deviation of 50KHz. A test signal of 4V, 2KHz produces a frequency deviation of 16KHz.
   Determine the modulation index ?

$$m = \frac{\Delta f}{f_m} = \frac{16k}{2k} = 8$$

4. A FM system has the following properties: $f_m$ = 3KHz, $V_m$ = 3v, m = 4, $V_c$ = 2v, $f_c$ = 200KHz

   Determine the frequency deviation ?

   Modulation index $$m = \frac{\Delta f}{f_m}$$

   Thus $$\Delta f = m f_m = (4)(3K) = 12KHz$$

5. A FM system has the following properties: $f_m$ = 3KHz, $V_m$ = 3v, m = 4, $V_c$ = 2v, $f_c$ = 200KHz

Determine the modulation sensitivity ?

Modulation index $$m = \frac{\Delta\mu}{\mu_m} = \frac{k_f V_m}{\mu_m}$$

Thus $$\Delta\mu = k_f V_m$$

Thus $$k_f = \frac{\Delta\mu}{V_m} = 25132$$

6. Given that the deviation constant is 1kHz/10mV, what is the shift in frequency for a voltage level of 50 mV?

Frequency deviation Δf = $50 \times \frac{1}{10} = 5\text{kHz}$

7. In an FM system when the audio frequency is 300 Hz and the audio voltage is 2.0V, the deviation is 5kHz. If the audio voltage is now increased to 6V what is the new deviation? If the voltage is now increased to 9V and the frequency dropped to 100Hz what is the deviation? Find the modulation index in each case.

7. In an FM system when the audio frequency is 300 Hz and the audio voltage is 2.0V, the deviation is 5kHz. If the audio voltage is now increased to 6V what is the new deviation? If the voltage is now increased to 9V and the frequency dropped to 100Hz what is the deviation? Find the modulation index in each case.

**$\delta = \Delta f$ = Deviasi frekuensi maksimum**

$$¢ \propto V_m$$

$$\frac{¢}{V_m} = \frac{5}{2} = 2.5kHz/V$$

$$when \ \mathrm{V} = 6\mathrm{v}$$

$$¢ = 2.5 \times 6 = 15\mathrm{kHz}$$

$$when \ \mathrm{V} = 9\mathrm{v}$$

$$¢ = 2.5 \times 9 = 22.5\mathrm{kHz}$$

$$m_f = \frac{¢}{f_m} = \frac{5}{0.3} = 16.67$$

$$m_f = \frac{¢}{f_m} = \frac{15}{0.3} = 50$$

$$m_f = \frac{¢}{f_m} = \frac{22.5}{0.1} = 225$$

8. Find the carrier and modulating frequencies, the modulating index, and the max. deviation of an FM wave below. What power will the wave dissipate in a 10 ohm resistor?

$$v = 12\sin\left(6\times10^{8}t + 5\sin 1250t\right)$$

Compare this with:

$$v = A\sin\left(\mu_c t + \frac{¢}{f_m}\sin\mu_m t\right)$$

$$f_c = \frac{\mu_c}{2®} = \frac{6\times10^{8}}{2®} = 95.5\text{MHz} \qquad f_m = \frac{\mu_m}{2®} = \frac{1250}{2®} = 199\text{Hz}$$

Modulating index=5 as given.

Power, $$P = \frac{V^2{}_{rms}}{R} = \frac{\left(12/\sqrt{2}\right)^2}{10} = \frac{72}{10} = 7.2W$$